

MATH3303 – Lecture 33

Counting Roots; Irreducibility in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$

Review Notes

Reference: Lee, Abstract Algebra, Sections 11.1–11.2

Overview

This lecture:

- Bounds the number of roots of a polynomial by its degree.
- Uses this to detect irreducibility in $\mathbb{Z}_p[x]$ (reduction-mod- p trick).
- Begins the systematic study of irreducibility in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$: Rational Roots Theorem, content and primitive polynomials, Gauss's Lemma, and the bridge from $\mathbb{Q}[x]$ -reducibility back to $\mathbb{Z}[x]$ -reducibility.

1 Counting Roots

Lemma 1.1: Degree bounds the number of roots (from lecture; cf. Lee, Cor. 11.3)

Let K be a field and $f(x) \in K[x]$ a nonzero polynomial. If $\deg(f(x)) = n$, then $f(x)$ has at most n roots in K .

Proof. By induction on n .

Base case $n = 0$: $f(x)$ is a nonzero constant, which has no roots.

Inductive step: Assume the result holds for degree n , and let $\deg(f(x)) = n + 1$.

If $f(x)$ has no roots, we are done. Otherwise, let a be a root of $f(x)$. By the Factor Theorem (Lecture 32),

$$f(x) = (x - a)g(x) \quad \text{for some } g(x) \in K[x].$$

Moreover, $\deg(g(x)) = n$, so by the inductive hypothesis $g(x)$ has at most n roots in K .

Let b be any root of $f(x)$. Then

$$f(b) = (b - a)g(b) = 0,$$

so either $b = a$ or $g(b) = 0$. Hence b is either the root a or one of the at most n roots of $g(x)$. Therefore $f(x)$ has at most $n + 1$ roots in K . $\square$

Example 1.1: $f(x) = x^3 + x + 1$ is irreducible in $\mathbb{Z}_5[x]$ (from lecture)

Suppose for contradiction f were reducible: $f(x) = g(x)h(x)$. WLOG $\deg(g) = 1$ and $\deg(h) = 2$. But $\deg(g) = 1$ means $f(x)$ has a root in $\mathbb{Z}_5$.

Check all elements of $\mathbb{Z}_5$:

$$\begin{aligned} f(0) &= 0 + 0 + 1 = 1, \\ f(1) &= 1 + 1 + 1 = 3, \\ f(2) &= 8 + 2 + 1 = 11 \equiv 1 \pmod{5}, \\ f(3) &= 27 + 3 + 1 = 31 \equiv 1 \pmod{5}, \\ f(4) &\equiv 1 \cdot 4 + 4 + 1 = 9 \equiv 4 \pmod{5}. \end{aligned}$$

No roots in $\mathbb{Z}_5$. Contradiction. Hence $f(x)$ is irreducible in $\mathbb{Z}_5[x]$.

Example 1.2: $f(x) = 10x^{2026} + x^3 + 6x - 4$ has no roots in $\mathbb{Z}$ (from lecture)

Consider the natural ring map $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_5$.

Suppose for contradiction $f(a) = 0$ for some $a \in \mathbb{Z}$. Then $\varphi(f(a)) = 0$ in $\mathbb{Z}_5$. But

$$\varphi(f(a)) \equiv 10a^{2026} + a^3 + 6a - 4 \equiv a^3 + a + 1 \pmod{5}.$$

So $a \pmod{5}$ would be a root of $x^3 + x + 1$ in $\mathbb{Z}_5$. But that polynomial has no roots in $\mathbb{Z}_5$ by the previous example. Contradiction.

Therefore $f(x)$ has no integer roots.

2 Irreducible Polynomials in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$

Theorem 2.1: Rational Roots Theorem (from lecture; cf. Lee, Thm. 11.3)

Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$ with $a_n \neq 0$. Suppose $q \in \mathbb{Q}$ is a root of $f(x)$. If $q = \frac{r}{s}$ with $r, s \in \mathbb{Z}$, $s \neq 0$, and $\gcd(r, s) = 1$, then

$$r \mid a_0 \quad \text{and} \quad s \mid a_n.$$

Proof. We have

$$0 = f(q) = f\left(\frac{r}{s}\right) = a_0 + a_1\frac{r}{s} + \cdots + a_n\frac{r^n}{s^n}.$$

Multiplying by s^n :

$$a_0s^n + a_1rs^{n-1} + \cdots + a_{n-1}r^{n-1}s + a_nr^n = 0.$$

Rearranging:

$$a_0s^n = -r(a_1s^{n-1} + \cdots + a_nr^{n-1}).$$

So $r \mid a_0s^n$. Since $\gcd(r, s) = 1$, we have $r \mid a_0$.

Similarly, $s \mid a_nr^n$, and since $\gcd(r, s) = 1$, $s \mid a_n$. □

Example 2.1: $f(x) = 3x^3 + 2x^2 - 2x - 8$ (from lecture; cf. Lee, Example 11.6)

By the Rational Roots Theorem, the only possible rational roots are

$$\pm 1, \pm 2, \pm 4, \pm 8, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{4}{3}, \pm \frac{8}{3}.$$

Trying them all, we see the only rational root is $\frac{4}{3}$.

By polynomial division,

$$f(x) = \left(x - \frac{4}{3}\right)(3x^2 + 6x + 6) = (3x - 4)(x^2 + 2x + 2).$$

Both factors are irreducible over $\mathbb{Z}$. For $3x - 4$, this is clear (degree 1 with content 1). For $x^2 + 2x + 2$:

- If it were reducible, then $x^2 + 2x + 2 = g(x)h(x)$ with $\deg(g) = \deg(h) = 1$.
- Hence it would have a root in $\mathbb{Z}$.
- But $x^2 + 2x + 2 = (x + 1)^2 + 1 \geq 1$ for all $x \in \mathbb{Z}$. No integer roots. Contradiction.

Towards a convenient criterion for irreducibility in $\mathbb{Q}[x]$. It is actually enough to study irreducibility of *primitive* polynomials in $\mathbb{Z}[x]$.

Definition 2.1: Content; primitive polynomial (from lecture; cf. Lee, Def. 11.2)

Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$ be nonzero. The *content* of $f(x)$ is

$$c(f) := \gcd(a_0, a_1, \dots, a_n),$$

the largest positive integer dividing all coefficients of $f(x)$.

$f(x)$ is called *primitive* if $c(f) = 1$.

Lemma 2.1: Gauss's Lemma (from lecture; cf. Lee, Lemma 11.2)

If $f(x), g(x) \in \mathbb{Z}[x]$ are primitive, then $f(x)g(x)$ is primitive.

Proof. Assume for contradiction $f(x)g(x)$ is not primitive: some prime p divides every coefficient of $f(x)g(x)$.

Let $\pi : \mathbb{Z} \rightarrow \mathbb{Z}_p$ be the natural ring map (reduction mod p). It extends to the ring map

$$\bar{\pi} : \mathbb{Z}[x] \longrightarrow \mathbb{Z}_p[x]$$

by reducing each coefficient mod p . Note: $\mathbb{Z}_p$ is a field, so $\mathbb{Z}_p[x]$ is an integral domain.

Now,

$$\bar{\pi}(fg) = \bar{\pi}(f)\bar{\pi}(g).$$

LHS: Since p divides every coefficient of fg , $\bar{\pi}(fg) = 0$ in $\mathbb{Z}_p[x]$.

RHS: f is primitive, so some coefficient of f is not divisible by p , hence $\bar{\pi}(f) \neq 0$. Similarly $\bar{\pi}(g) \neq 0$.

So $\bar{\pi}(f)\bar{\pi}(g) = 0$ with both factors nonzero in $\mathbb{Z}_p[x]$, contradicting the fact that $\mathbb{Z}_p[x]$ is an integral domain. $\square$

3 From $\mathbb{Q}[x]$ -Reducibility back to $\mathbb{Z}[x]$ -Reducibility

Claim. If a polynomial in $\mathbb{Z}[x]$ is reducible in $\mathbb{Q}[x]$, then it is reducible in $\mathbb{Z}[x]$ as well.

Remark 3.1: Warning (from lecture)

This is *not* true in general for the chain $R[x] \subseteq K[x]$ with $K = \text{Frac}(R)$ – the analogous claim can fail if we are not careful about constants. The statement below is the precise version.

Theorem 3.1: Reducible in $\mathbb{Q}[x] \Rightarrow$ scaling lands in $\mathbb{Z}[x]$ (from lecture; cf. Lee, Thm. 11.4)

Let $f(x) \in \mathbb{Z}[x]$ and suppose $f(x) = g(x)h(x)$ for some $g(x), h(x) \in \mathbb{Q}[x]$. Then there exists a positive rational number $q \in \mathbb{Q}$ such that

$$qg(x) \in \mathbb{Z}[x] \quad \text{and} \quad \frac{1}{q}h(x) \in \mathbb{Z}[x].$$

The proof is given in Lecture 34.

Checklist of Skills

- ☐ Prove that a polynomial of degree n over a field has at most n roots, by induction.
- ☐ Use the no-roots test to detect irreducibility of cubics in $\mathbb{Z}_p[x]$.
- ☐ Apply the reduction-mod- p trick to rule out integer roots of a polynomial in $\mathbb{Z}[x]$.
- ☐ State and prove the Rational Roots Theorem.
- ☐ Apply the Rational Roots Theorem to a cubic and factor it over $\mathbb{Z}$.
- ☐ Define content of a polynomial and primitive polynomial.
- ☐ State and prove Gauss's Lemma using reduction mod p .
- ☐ State the theorem: reducibility in $\mathbb{Q}[x]$ for $f \in \mathbb{Z}[x]$ yields a scaling into $\mathbb{Z}[x]$.